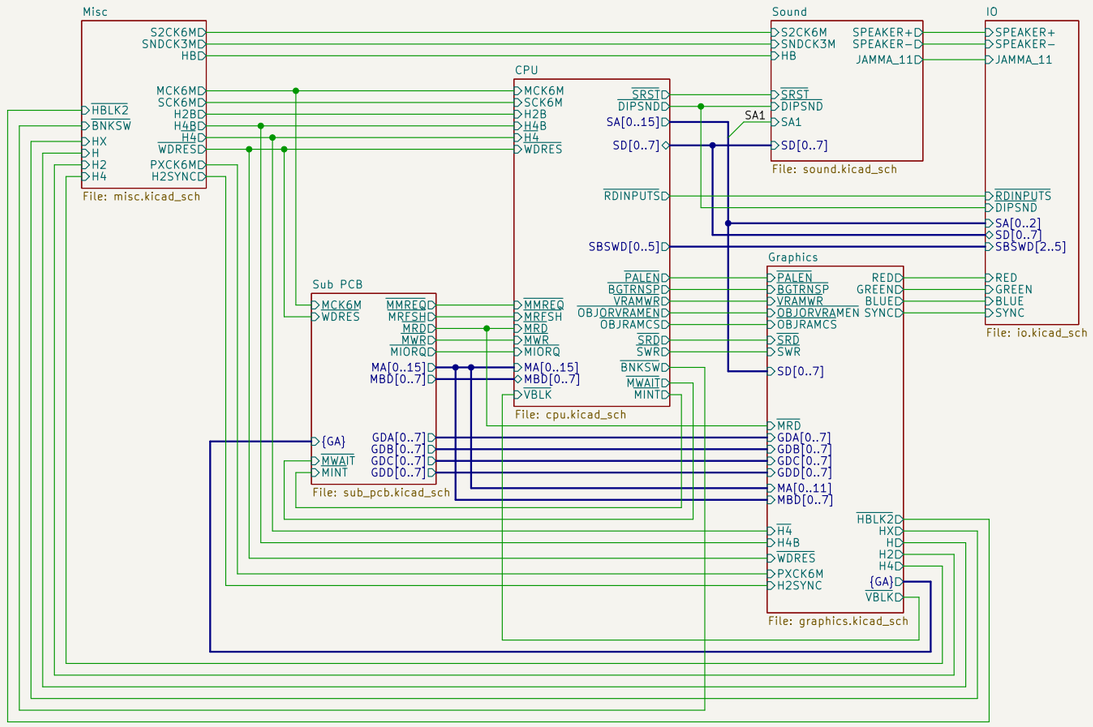


Note: The reference designator refers to the position on the PCB. If the reference designator already exists the letter is doubled. I.e. C1 -> CC1.

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Sheet: /		
File: tnzs.kicad_sch		
Title: The NewZealand Story – P0-043A		
Size: A3	Date: 2024-04-07	**Rev:**
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IC5
74LS174
MD0 MD1 MD2 MD3 MD4 MD5
D0 D1 D2 D3 D4 D5
Q0 Q1 Q2 Q3 Q4 Q5
Cp
hex flip flop
Mr
VCC GND

IC1
74F245
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
A0 A1 A2 A3 A4 A5 A6 A7
buffers
A->B
MRD CE
B0 B1 B2 B3 B4 B5 B6 B7
MBD0 MBD1 MBD2 MBD3 MBD4 MBD5 MBD6 MBD7
MBD[0..7]

MA11 MA12 MA13 MA14 MA15 MCK6M MBD4 MBD3 MBD5 MBD6 MBD2 MBD7 MBD0 MBD1 MINT MBUSRQ MHALT MMREQ MIORQ
JS2
Conn_01x20
VCC GND

MA10 MA9 MA8 MA7 MA6 MA5 MA4 MA3 MA2 MA1 MA0 MRFSH MM1 WDRES MBUSRQ MWAIT MBUSACK MWR MRD
JS1
Conn_01x20

NC: MM1, MBUSRQ, MHALT, MBUSACK

IC6
PAL16L8'
MA10 MA11 MA12 MA13 MA14 MA15 MWR MRD MMREQ GROMENB MDC2 MDC1 MDC0 MA9 MRAMCS
I1 I2 I3 I4 I5 I6 I7 I8 I9 I11
I/O13 I/O14 I/O15 I/O16 I/O17 I/O18 O12 O19
VCC VSS
GND

/MRAMCS = /MA14 & MA15 & /MDC2 & /MDC1
o13 = /MA14 & MA15 & /MDC2 & /MDC1
/GROMENB = GROMEN
/o19 = MA9 & MA10 & /MA11 & MA12 & MA14 & MA15 & /MWR & /MMREQ

MA[0..15]

IC3
Z80CPU
WDRES MCK6M VCC MINT MWAITD MM1 MRFSH MHALT MRD MWR MMREQ MIORQ MBUSRQ MBUSACK
RESET CLK NMI INT M1 RFSH WAIT HALT RD WR MREQ IORQ BUSRQ BUSACK
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
D0 D1 D2 D3 D4 D5 D6 D7
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MA14 MA15
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
GND

IC4C
74HC08
WDRES MCK6M VCC MINT

IC2
SRM20256
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MDC0
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
CS OE WE
MRD MWR
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
VCC GND

MD[0..7]

IC7 IC9 IC11 IC12
27C010
GA0 GA1 GA2 H8R GA4 GA5 GA6 GA7 GA8 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16
D0 D1 D2 D3 D4 D5 D6 D7
VCC PGM VPP CE OE GND
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7

IC8 IC10 IC13 IC14
27C010
GA0 GA1 GA2 H8R GA4 GA5 GA6 GA7 GA8 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16
D0 D1 D2 D3 D4 D5 D6 D7
VCC PGM VPP CE OE GND
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7

GROMEN
GROMENB
{GA}

JS6 JS5 JS4 JS3
Conn_01x20
GND GDB0 GDB1 GDB2 GDB3 GDB7 GDB6 GDB5 GDB4 VCC GA8 GA7 GA6 GA5 GA4 H8R GA2 GA1 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16 GROMEN GA0 GDA7 GDA6 GDA5 GDA4

GDA[0..7]
GDB[0..7]
GDD[0..7]
GDC[0..7]

JS10 JS9 JS8 JS7
Conn_01x20
GND GDD0 GDD1 GDD2 GDD3 GDD7 GDD6 GDD5 GDD4 VCC GDC0 GDC1 GDC2 GDC3 GDC7 GDC6 GDC5 GDC4

IC4D
74HC08
GND GND

IC4B
74HC08
GND GND

IC4A
74HC08
GND GND

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Sheet: /Sub PCB/
File: sub_pcb.kicad_sch
Title: The NewZealand Story – P0-043A
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MBD[0..7]
MA[0..15]

B06-101
PAL16L8'
VCC
GND
MA15 MA14 MA13 MA12 MA11 MA10 MA9
MMREQ MRFSH MWR
VRAMWR BGTRNSP MSHRAMRQ OBJORVRAMEN OBJORVRAMEN

CC2
74HC174
MBD5 MBD4 MBD3 MBD2 MBD1 MBD0
SRST
WDRES
BNKSW
hex flip flop

B3
74F157
S Quad Mux

B53-24.U1
27C010
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13
MBD0 MBD1 MBD2 MBD3 MBD4 MBD5 MBD6 MBD7
PGM VPP CE OE

D10C
74LS04
B4A
74F32
B4B
74F32
CC5A
74F08
JR1
JR2

A3B
74LS107
VCC VBLK GND
MIORQ
MINT

H9A
74LS20
MA15 MA14
MA13
D10A
74LS04
E7A
74LS08
OBJRAMCS
H4D
H9B
74LS20

E10B
74LS139
decoder
MMREQ MRFSH GND

H2B H4B
E7B
74LS08
MCK6MD
E9A
74LS74
OBJORVRAMEN SSHRAMEN MSHRAMRQ
D5A CC4A
74LS32
CC5B
74F08
MWAIT

SSHRAMRQ
D5B
74LS32
SWAIT
MSHRAMEN
MSHRAMRQ
CC4B
74LS32
E6F
74F04
E6E
74F04
SSHRAMRQ
CC4C
74LS32
SSHRAMEN

MWR MSHRAMEN
D5C
74LS32
OBJORVRAMEN SSHRAMEN
CC4D
74LS32
CC5C
74F08
MSHRAMEN SSHRAMEN
CC5D
74F08

F5
uPD4464
R/W
CS1 CS2 OE
SHA12 SHA11 SHA10 SHA9 SHA8 SHA7 SHA6 SHA5 SHA4 SHA3 SHA2 SHA1 SHA0
SHD7 SHD6 SHD5 SHD4 SHD3 SHD2 SHD1 SHD0

SA0 SRD GND GND MA0 MRD GND GND SSHRAMEN GND
F4
74LS157
SHA0 OE SHA[0..12]

SA8 SA7 SA6 SA5 MA8 MA7 MA6 MA5 SSHRAMEN GND
F3
74LS157
SHA8 SHA7 SHA6 SHA5

SA4 SA3 SA2 SA1 MA4 MA3 MA2 MA1 SSHRAMEN GND
E3
74LS157
SHA4 SHA3 SHA2 SHA1

SA12 SA11 SA10 SA9 MA12 MA11 MA10 MA9 SSHRAMEN GND
E2
74LS157
SHA12 SHA11 SHA10 SHA9

D3
74HC245
MBD7 MBD6 MBD5 MBD4 MBD3 MBD2 MBD1 MBD0
SHD7 SHD6 SHD5 SHD4 SHD3 SHD2 SHD1 SHD0
MRD MSHRAMEN
buffers
A->B

D4
74HC245
SD7 SD6 SD5 SD4 SD3 SD2 SD1 SD0
SHD7 SHD6 SHD5 SHD4 SHD3 SHD2 SHD1 SHD0
SRD SSHRAMEN
buffers
A->B

SHD[0..7]

Pin 12: 0xF800..0xFFFF: NC
Pin 13: 0xF000..0xF2FF: spritegen

Pin 16: 0xC000..0xCFFF: OBJCODELO
Pin 17: 0xD000..0xDFFF: OBJCODEHI
Pin 18: 0xE000..0xEFFF: /MSHRAMRQ
Pin 19: 0xC000..0xDFFF or 0xF000..0xF5FF: OBJORVRAMEN

PALEN (NC) = MA15 & MA14 & MA13 & MA12 & MA11 & MMREQ & MRFSH
VRAMWR = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MMREQ & MRFSH
BGTRNSP = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MA9 & MMREQ & MRFSH & MWR
BNKSW = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MA9 & MMREQ & MRFSH & MWR
OBJCODELO (NC)= MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
OBJCODEHI (NC)= MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
MSHRAMRQ = MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
OBJORVRAMEN = MMREQ & MRFSH & MA15 & MA14 & (MA13 & MA12 & MA11 & MA9 + MA13 & MA12 & MA11 & MA10 + MA13)

Pin 12: 0x0000..0x7FFF or 0x8000..0x9FFF: SROMCS
Pin 13: 0xE000..0xEFFF or 0xD000..0xDFFF: SSHRAMRQ
Pin 14: 0xA000..0xAFFF: SBNKSW
Pin 15: 0xB000..0xBFFF: DIPSND
Pin 16: 0xC000..0xCFFF: RDINPUTS
Pin 17: 0xF000..0xFFFF: PALEN

SROMCS = (SA14 & SA13 + SA15) & SMREQ & SRFSH
SSHRAMRQ = (SA13 & SA12 + SA13 & SA12) & SA15 & SA14 & SMREQ & SRFSH
SBNKSW = SA15 & SA14 & SA13 & SA12 & SWR & SMREQ & SRFSH
DIPSND = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
RDINPUTS = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
PALEN = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
SRA13 = SA15 & SBSWD0 + SA15 & SA13
SRA14 = SA15 & SBSWD1 + SA15 & SA14

SA[0..15]

U3
Z80CPU
VCC GND
SRST SCK6MD VCC VCC SRFSH SWAIT
RESET CLK NMI INT M1 RFSH WAIT HALT
SRD SWR SMREQ MMREQ IORQ RD WR
VCC BUSRQ BUSACK
SWR SRD
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
D0 D1 D2 D3 D4 D5 D6 D7
SA0 SA1 SA2 SA3 SA4 SA5 SA6 SA7 SA8 SA9 SA10 SA11 SA12 SA13 SA14 SA15
SD0 SD1 SD2 SD3 SD4 SD5 SD6 SD7

A3A
74LS107
VCC VBLK GND

B06-13
PAL16L8'
SA15 SA14 SA13 SA12 SA11 SWR SMREQ SRFSH SBSWD1 SBSWD0
SSHRAMRQ SBNKSW DIPSND RDINPUTS PALEN SRA13 SROMCS SRA14
GND

SRA[13..14]
SBSWD[0..5]

H5
74HC174
SD5 SD4 SD3 SD2 SD1 SD0
SBSWD5 SBSWD4 SBSWD3 SBSWD2 SBSWD1 SBSWD0
SBNKSW SRST
hex flip flop

SD[0..7]

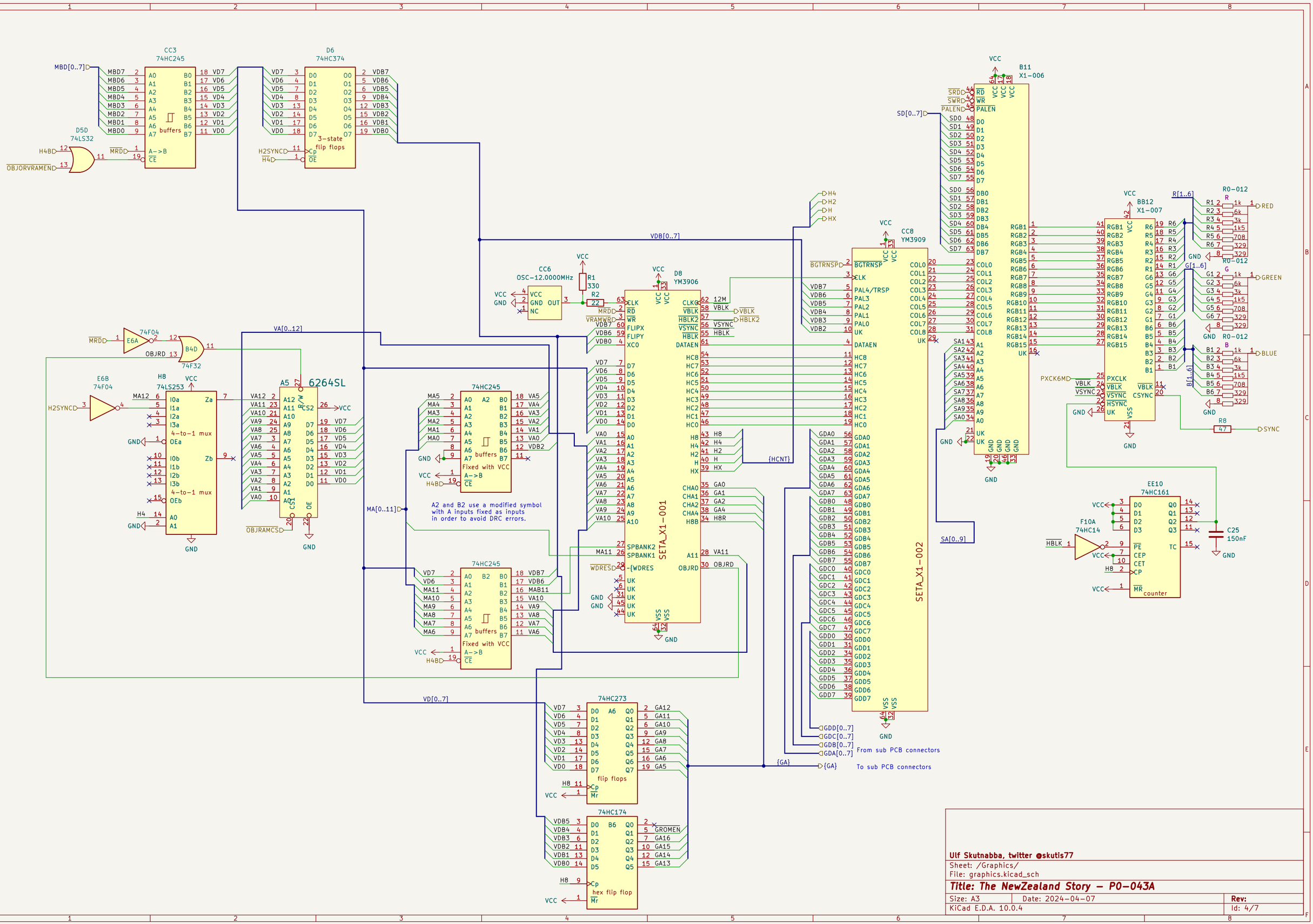
B53-25.U3
27512
SA0 SA1 SA2 SA3 SA4 SA5 SA6 SA7 SA8 SA9 SA10 SA11 SA12 SRA13 SRA14 SA15
SD0 SD1 SD2 SD3 SD4 SD5 SD6 SD7
GND GND

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Sheet: /CPU/
File: cpu.kicad_sch

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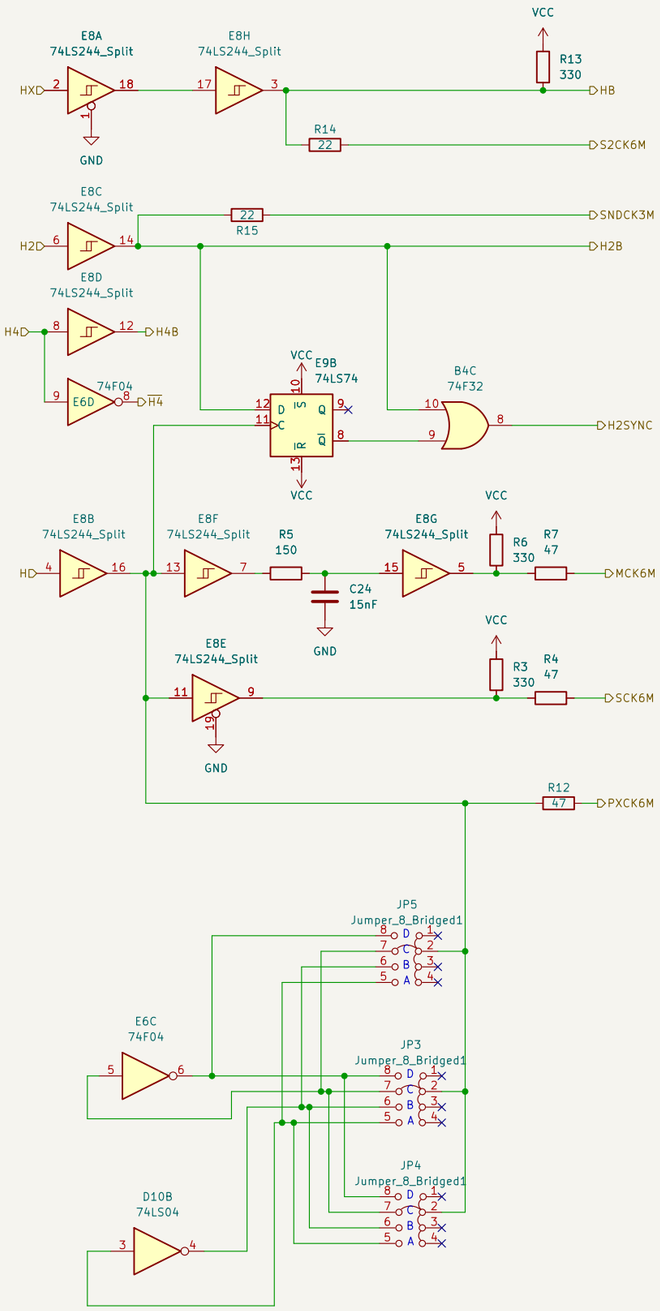


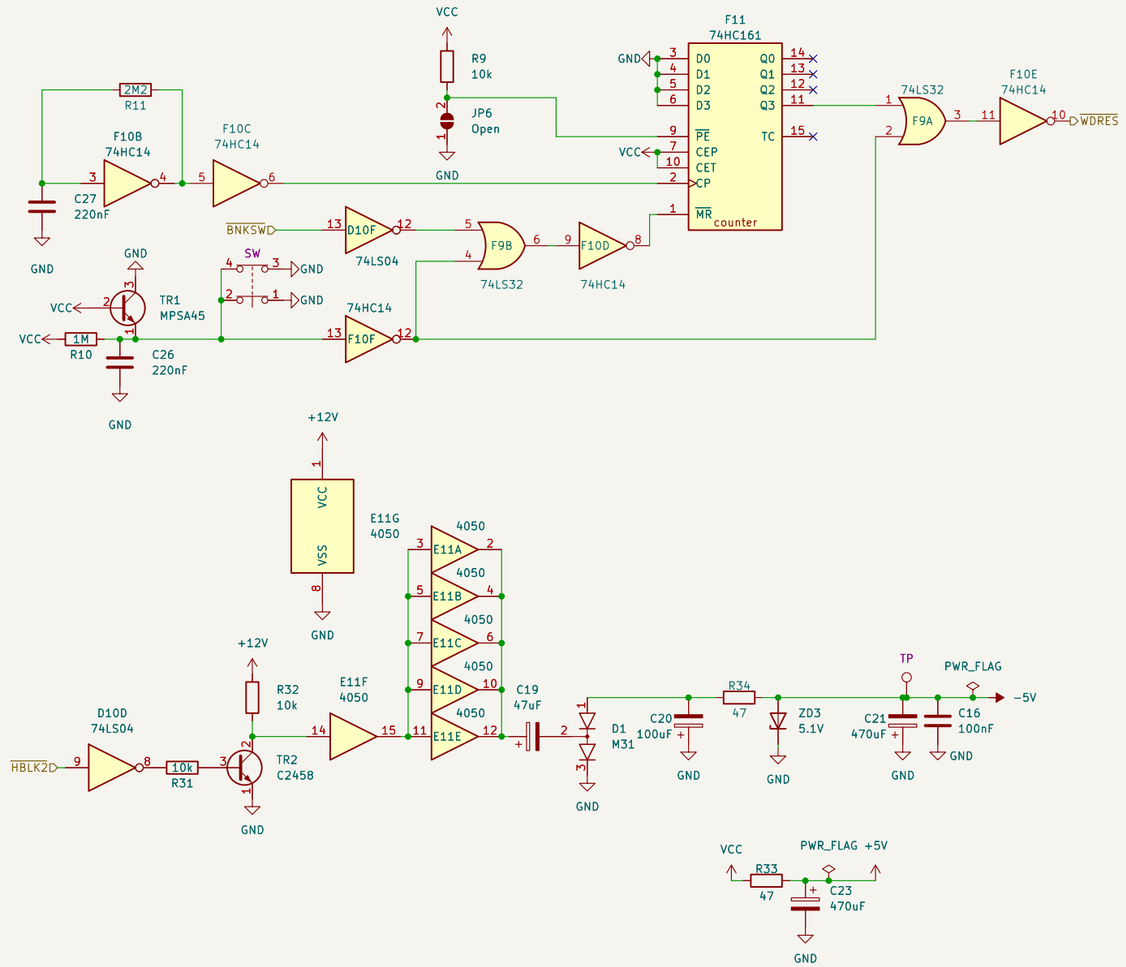
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Sheet: /Graphics/
File: graphics.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	**Rev:**
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Sheet: /Misc/
File: misc.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	**Rev:**
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RA3
2.2K x 8
1P_START 2
1P_BTN_3 3
1P_BTN_2 4
1P_BTN_1 5
1P_DOWN 6
1P_UP 7
1P_RIGHT 8
1P_LEFT 9
VCC

RA4
2.2K x 8
2P_START 2
2P_BTN_3 3
2P_BTN_2 4
2P_BTN_1 5
2P_DOWN 6
2P_UP 7
2P_RIGHT 8
2P_LEFT 9
VCC

RA5
2.2k x 4
TILT 2
SERVICE 3
COIN_2 4
COIN_1 5
VCC

JAMMA_CONN
PWR_FLAG
GND
PWR_FLAG
VCC
+12V
ZD2
> 12.0V
C4
220uF
GND
GND
COIN_CNT_1 8
COIN_LOCK_1 9
SPEAKER+ 10
JAMMA_11D 11
REDD 12
BLUED 13
GND 15
COIN_1 16
1P_START 17
1P_UP 18
1P_DOWN 19
1P_LEFT 20
1P_RIGHT 21
1P_BTN_1 22
1P_BTN_2 23
1P_BTN_3 24
25
26
27
28
GND
1 GND
2 VSSA
3 VDDA
4 +5V
5 −5V
6 +12V
COIN_CNT_1
LOCK_COIL_1
SPEAKER_(+)
NOT_USED
RED
BLUE
VIDEO_GND
TEST
COIN_1
1P_START
1P_UP
1P_DOWN
1P_LEFT
1P_RIGHT
1P_BTN_1
1P_BTN_2
1P_BTN_3
1P_BTN_4
1P_SELECT
GND
GND
A GND
B GND
C +5V
D +5V
E −5V
F +12V
COIN_CNT_2
LOCK_COIL_2
SPEAKER_(−)
NOT_USED
GREEN
SYNC
SERVICE
TILT
COIN_2
2P_START
2P_UP
2P_DOWN
2P_LEFT
2P_RIGHT
2P_BTN_1
2P_BTN_2
2P_BTN_3
2P_BTN_4
2P_SELECT
GND
GND
J COIN_CNT_2
K COIN_LOCK_2
L
M SPEAKER−
N GREEN
P SYNC
R SERVICE
S TILT
T COIN_2
U 2P_START
V 2P_UP
W 2P_DOWN
X 2P_LEFT
Y 2P_RIGHT
Z 2P_BTN_1
a 2P_BTN_2
b 2P_BTN_3
c
d
e
f GND
PWR_FLAG
C1
470uF
+12V
GND GND
ZD1
> 5.0V
VCC

E12 X2-004
1P_START_B
1P_BTN_3_B
1P_BTN_2_B
1P_BTN_1_B
1P_DOWN_B
1P_UP_B
1P_START_B
1P_BTN_3_B
1P_BTN_2_B
1P_BTN_1_B
1P_DOWN_B
1P_UP_B

D12 X2-004
1P_RIGHT_B
1P_LEFT_B
2P_START_B
2P_BTN_3_B
2P_BTN_2_B
2P_BTN_1_B
1P_RIGHT_B
1P_LEFT_B
2P_START_B
2P_BTN_3_B
2P_BTN_2_B
2P_BTN_1_B

CC12 X2-004
2P_DOWN_B
2P_UP_B
2P_RIGHT_B
2P_LEFT_B
TILT_B
SERVICE_B
2P_DOWN_B
2P_UP_B
2P_RIGHT_B
2P_LEFT_B
TILT_B
SERVICE_B

B12 X2-004
COIN_2_B
COIN_1_B
COIN_2_B
COIN_1_B

RA2
8 x 10k
VCC
SWA1
SWA2
SWA3
SWA4
SWA5
SWA6
SWA7
SWA8

SWA
SW_DIP_x08
GND
SWA1
SWA2
SWA3
SWA4
SWA5
SWA6
SWA7
SWA8

RA1
8 x 10k
VCC
SWB1
SWB2
SWB3
SWB4
SWB5
SWB6
SWB7
SWB8

SWB
SW_DIP_x08
GND
SWB1
SWB2
SWB3
SWB4
SWB5
SWB6
SWB7
SWB8

SWA[1..8]
SWB[1..8]

H3
74HC257
SWA1 2
SWB1 3
SWA2 5
SWB2 6
SWA4 14
SWB4 13
SWA3 11
SWB3 10
SA0 1
15
4 SD0
7 SD1
12 SD3
9 SD2

H2
74HC257
SWA5 2
SWB5 3
SWA6 5
SWB6 6
SWA8 14
SWB8 13
SWA7 11
SWB7 10
SA0 1
15
4 SD4
7 SD5
12 SD7
9 SD6

F9C
74LS32
SA2 9
10
8
DIPSNDD

SD[0..7]
SD[0..7]
SA[0..2]

E10A
74LS139
GND
SA1
RDINPUTSD
3
2
1
A1
A0
E
O0 4
O1 5
O2 6
O3 7
decoder

D11
X1-004
SA0 37
E3 41
E2 42
E1 40
SD1 35
SD0 34
SD5 33
SD4 32
1P_START_B 2
1P_BTN_3_B 3
1P_BTN_2_B 4
1P_BTN_1_B 5
1P_DOWN_B 6
1P_UP_B 7
1P_RIGHT_B 8
1P_LEFT_B 9
2P_START_B 10
2P_BTN_3_B 11
2P_BTN_2_B 12
2P_BTN_1_B 13
2P_DOWN_B 14
2P_UP_B 15
2P_RIGHT_B 16
2P_LEFT_B 17
COIN_LOCK_1_B 22
COIN_LOCK_2_B 23
COIN_CNT_2_B 24
COIN_CNT_1_B 25
38
36
43
GND
VCC
GND
39
42
IOD1
IOD2
IOD3
IOD4
INA1
INA2
INA3
INA4
INA5
INA6
INA7
INA8
INB1
INB2
INB3
INB4
INB5
INB6
INB7
INB8
IOA1
IOA2
IOA3
IOA4
UK
UK
UK
UK
UK
UK
IOC1 18 TILT_B
IOC2 19 SERVICE_B
IOC3 20 COIN_2_B
IOC4 21 COIN_1_B
D7 51 SD7
D6 50 SD6
D5 49 SD5
D4 48 SD4
D3 47 SD3
D2 46 SD2
D1 45 SD1
D0 44 SD0
IOB1 31 (SBSWD5)
IOB2 30 (SBSWD5)
IOB3 29 SBSWD3
IOB4 28 SBSWD2
VCC 52
VCC 27
GND 26
1
SBSWD[2..5]

The I/O direction of each port in X1−004 is
probably determined by connecting the pins
36,38,39, 42 and 43 to VCC and GND accordingly.

In tnzs IOD −> IOC, IOB −> IOA

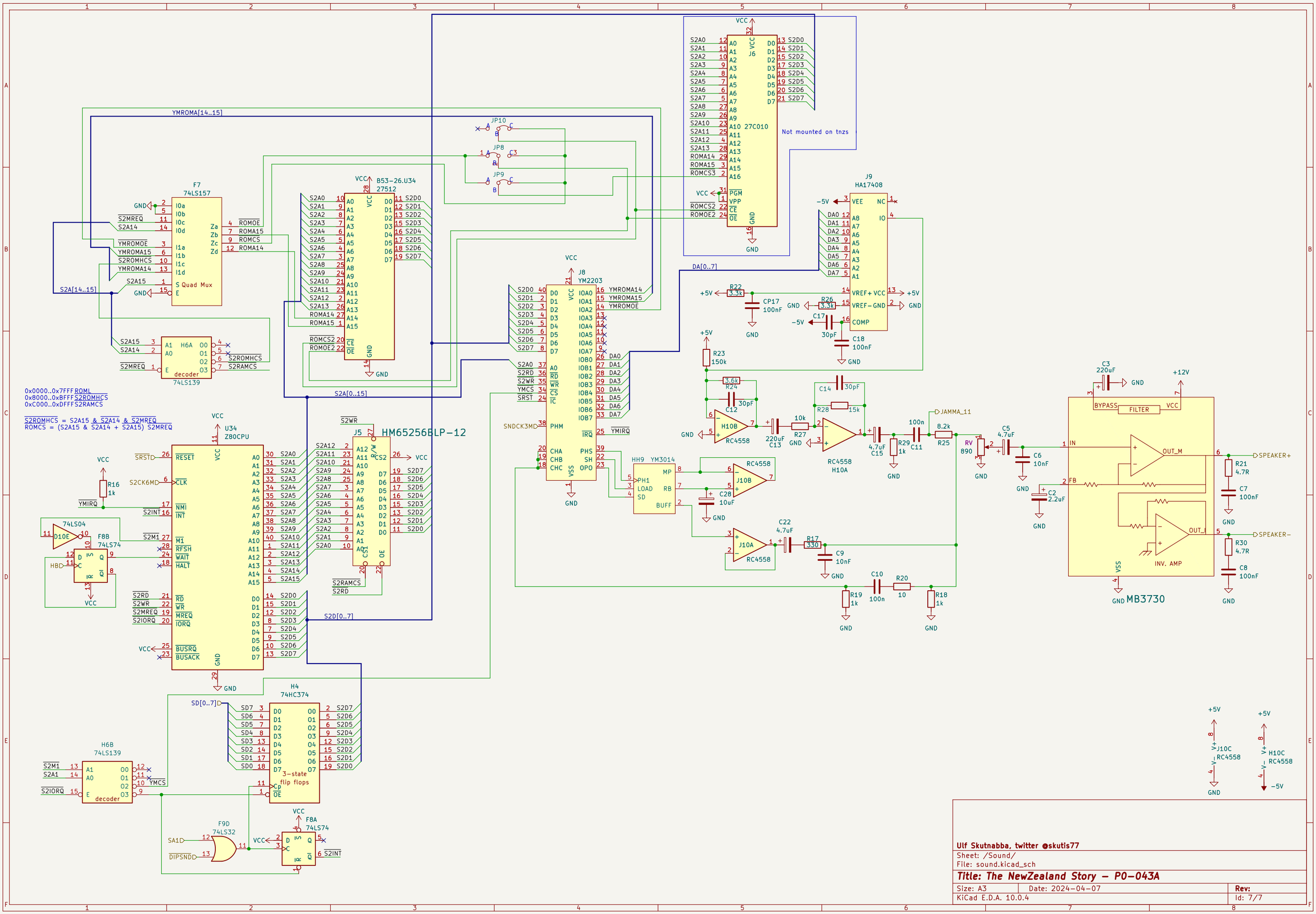
E1 E2 E3 Active output
H L L INA−> D, IOB −> IOA
H L H INB −> D, IOB −> IOA
L H L IOC −> IOD, IOB −> IOA
X X X IOB −> IOA

All inputs are active low, when the outputs
are not active they are in tri−state state.
Except for IOA in this case.

SBSWD4 JP7 SBSWD5

F12
TD62064
COIN_LOCK_2_B 3
COIN_CNT_2_B 6
COIN_CNT_1_B 11
COIN_LOCK_1_B 14
I1
I2
I3
I4
O1 2 COIN_LOCK_2
O2 7 COIN_CNT_2
O3 9 COIN_CNT_1
O4 16 COIN_LOCK_1
+12V 1
8
10
15
COM GND 4
COM GND 5
NC GND 12
NC GND 13
GND

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Sheet: /IO/
File: io.kicad_sch
Title: The NewZealand Story − P0−043A
Size: A3
Date: 2024−04−07
Rev:
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YMROMA[14..15]
S2A[14..15]
S2A[0..15]
DA[0..7]
S2D[0..7]
SD[0..7]
F7
74LS157
S Quad Mux
B53-26.U34
27512
J6
27C010
Not mounted on tnzs
J9
HA17408
J8
YM2203
JP10
JP8
JP9
H6A
decoder
74LS139
0x0000..0x7FFF ROML
0x8000..0xBFFF S2ROMHCS
0xC000..0xDFFF S2RAMCS
S2ROMHCS = S2A15 & S2A14 & S2MREQ
ROMCS = (S2A15 & S2A14 + S2A15) S2MREQ
U34
Z80CPU
J5
HM65256BLP-12
H6B
74LS139
H4
74HC374
3-state flip flops
F9D
74LS32
F8A
74LS74
F8B
74LS74
74LS04
HH9
YM3014
J10A
J10B
J10C
RC4558
H10A
H10B
H10C
MB3730
INV. AMP
JAMMA_11
SPEAKER+
SPEAKER-
Not mounted on tnzs
Ulf Skutnabba, twitter @skutis77
Sheet: /Sound/
File: sound.kicad_sch
Title: The NewZealand Story – P0-043A
Size: A3
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